# IGCSE 0607 Practice Questions — E7.2 — Combined Questions & Markscheme

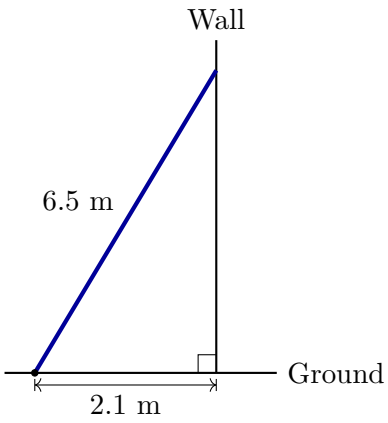
## Right-angled triangles

**Total Questions:** 9 — **Total Marks:** 62 Marks

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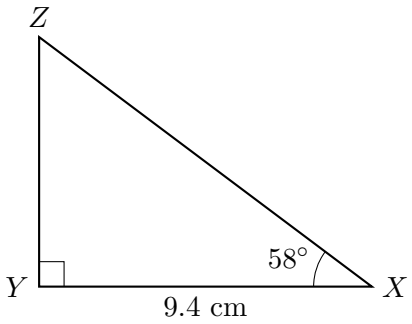

- **E7.2** Right-angled triangles
  → **Paper 4 (Calculator)** *(9 Qs)*

**1** A straight ladder of length 6.5 m leans against a vertical wall.
The base of the ladder is 2.1 m from the bottom of the wall on horizontal ground.




Calculate the angle the ladder makes with the horizontal ground.

..... ° **[3]**



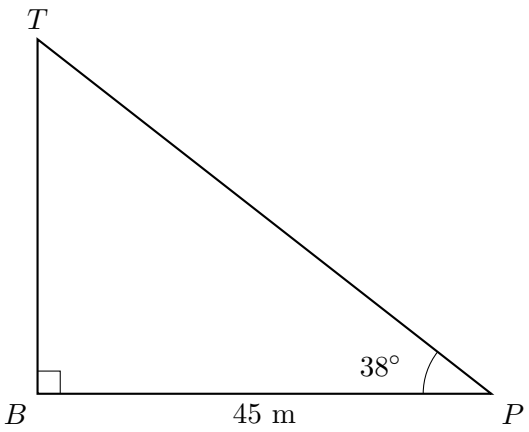

NOT TO SCALE

**2** Triangle $XYZ$ is right-angled at $Y$.
Angle $ZXY = 58°$ and $XY = 9.4$ cm.

Calculate the length of $XZ$.

..... cm **[3]**

**3** A surveyor stands at a point $P$, which is 45 m away from the base, $B$, of a vertical tower $TB$. The angle of elevation of the top of the tower, $T$, from $P$ is 38°.




**(a)** Calculate the height of the tower, $TB$.

..... $m$ **[2]**

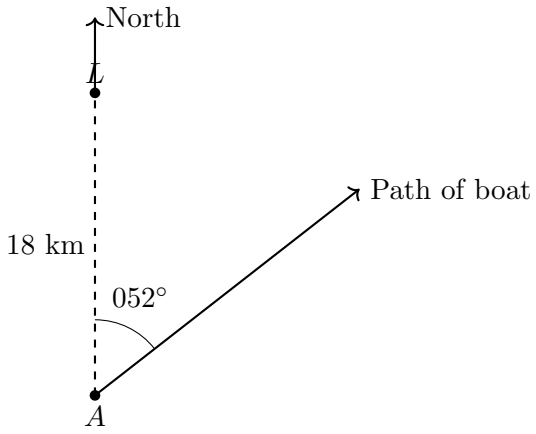
**(b)** Calculate the straight-line distance $PT$.

..... $m$ **[2]**

**(c)** A bird sits at a point $X$ on the tower, where $BX = 20$ m.
Calculate the angle of elevation of the bird from $P$.

..... ° **[3]**

**4** A boat sails from a point $A$ in a straight line on a bearing of 052°.
A lighthouse, $L$, is located 18 km due North of $A$.




NOT TO SCALE

**(a)** Calculate the shortest distance from the lighthouse $L$ to the path of the boat.

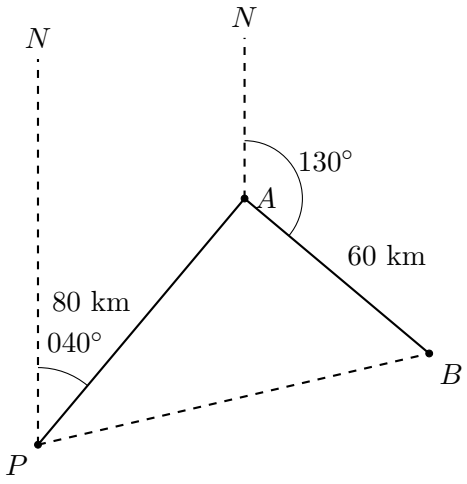
..... km **[2]**

**(b)** The boat travels at a constant speed of 14 km/h.

Calculate the time it takes for the boat to reach the point on its path closest to the lighthouse.
Give your answer in minutes, correct to the nearest minute.

..... minutes **[3]**

**5** A ship sails from port $P$ on a bearing of 040° for 80 km to point $A$.
It then changes course and sails on a bearing of 130° for 60 km to point $B$.




**(a)** Calculate the direct distance from $P$ to $B$.

$PB =$ ..... km **[3]**

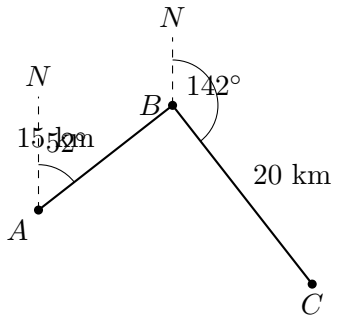
**(b)** Calculate the bearing of $B$ from $P$.

Bearing = ..... ° **[3]**

**(c)** Calculate the shortest distance from point $A$ to the path $PB$.

..... km **[3]**

**6** A ship sails from port $A$ to point $B$ on a bearing of 052° for 15 km.
It then sails from point $B$ to point $C$ on a bearing of 142° for 20 km.




**(a)** Show that angle $ABC = 90°$.

**[2]**

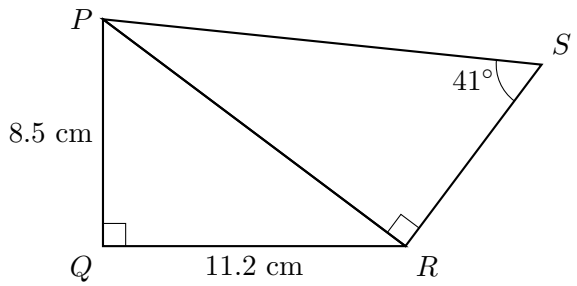
**(b)** Calculate the distance $AC$.

..... km **[3]**

**(c)** Calculate the bearing of $C$ from $A$.

..... ° **[4]**

**7** The diagram shows a quadrilateral $PQRS$ formed by two right-angled triangles, $PQR$ and $PRS$.
Angle $PQR = 90°$ and angle $PRS = 90°$.
$PQ = 8.5$ cm, $QR = 11.2$ cm and angle $PSR = 41°$.




NOT TO SCALE

**(a)** Calculate the length of $PR$.

..... cm **[2]**

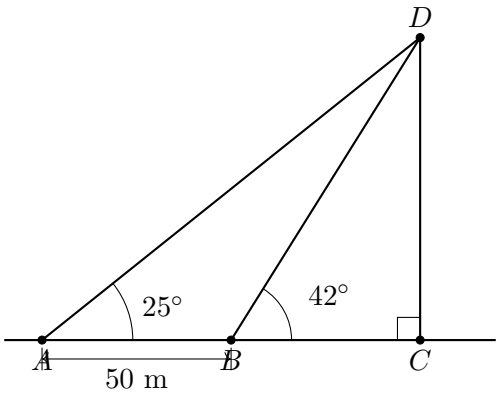
**(b)** Calculate the length of $PS$.

..... cm **[2]**

**(c)** Calculate the perpendicular distance from $R$ to the line $PS$.

..... cm **[2]**

**8** Point $A$ and point $B$ are on horizontal ground, in a straight line with the base $C$ of a vertical building $CD$.
The distance between $A$ and $B$ is 50 m.
The angle of elevation of the top of the building, $D$, from $A$ is 25° and from $B$ is 42°.



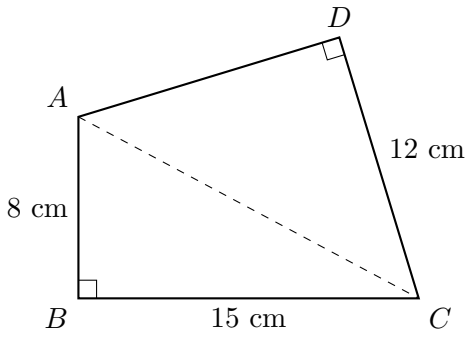

**(a)** Calculate the height of the building $CD$.

$CD =$ ..... $m$ **[5]**

**(b)** A bird flies directly along the straight path from $A$ to $D$.
Calculate the shortest distance from point $B$ to the flight path $AD$.

..... $m$ **[3]**

**9** The diagram shows a quadrilateral $ABCD$.
Angle $ABC = 90°$ and angle $ADC = 90°$.
$AB = 8$ cm, $BC = 15$ cm, and $CD = 12$ cm.




**(a)** Calculate the length of $AC$.

..... cm **[2]**

**(b)** Calculate the length of $AD$.

..... cm **[3]**

**(c)** Calculate the shortest distance from $D$ to the line $AC$.

..... cm **[3]**

**(d)** Calculate the size of angle $DAB$.

..... ° **[4]**

**1** Identifies cosine ratio: $\cos(\theta) = \dfrac{2.1}{6.5}$ **[M1]**

$\theta = \cos^{-1}\left(\dfrac{2.1}{6.5}\right)$ **[M1]**

71.2 **[A1]**

**[Back to Question]** *[Total Marks: 3]*

**2** $\cos(58°) = \dfrac{9.4}{XZ}$ **[M1]**

$XZ = \dfrac{9.4}{\cos(58°)}$ **[M1]**

17.7 or 17.73... **[A1]**

**[Back to Question]** *[Total Marks: 3]*

**3 (a)** $\tan(38°) = \dfrac{TB}{45}$ **[M1]**

35.2 (or 35.15 to 35.16) **[A1]**

**(b)** $\cos(38°) = \dfrac{45}{PT}$ or $PT^2 = 45^2 + 35.15^2$ **[M1]**

57.1 (or 57.10 to 57.11) **[A1]**

**(c)** $\tan(XPB) = \dfrac{20}{45}$ **[M1]**

Angle $= \tan^{-1}\left(\dfrac{20}{45}\right)$ **[M1]**

24.0 (or 23.96 . . . ) **[A1]**

**[Back to Question]** *[Total Marks: 7]*

**4 (a)** Uses $\sin(52°) = \dfrac{\text{distance}}{18}$ or $18 \times \sin(52°)$ **[M1]**

14.18... or 14.2 **[A1]**

**(b)** Distance to closest point $= 18 \times \cos(52°) = 11.082...$ km **[M1]**

Time $= \dfrac{11.082...}{14}$ $(= 0.7915...$ hours$)$ **[M1]**

$0.7915... \times 60 = 47.49... = 47$ **[A1]**

**[Back to Question]** *[Total Marks: 5]*

**5 (a)** Calculates $\angle PAB = 360° - (180° - 40°) - 130° = 90°$ (or similar valid reasoning to show right angle) **[M1]**

Uses Pythagoras' theorem: $PB^2 = 80^2 + 60^2$ **[M1]**

100 **[A1]**

**(b)** Uses trigonometry to find $\angle APB$: $\tan(APB) = \dfrac{60}{80}$ **[M1]**

$\angle APB = \tan^{-1}(0.75) = 36.869...^\circ$ **[A1]**

Bearing $= 040^\circ + 36.869...^\circ = 076.9$ (accept 077) **[A1]**

**(c)** Recognises the shortest distance $d$ is the perpendicular height of $\triangle PAB$ from $A$ to $PB$. **[M1]**

Uses area equation: $\dfrac{1}{2} \times 100 \times d = \dfrac{1}{2} \times 80 \times 60$ OR uses trigonometry $80\sin(53.13^\circ)$ or $60\sin(36.87^\circ)$ **[M1]**

48 **[A1]**

[Back to Question] *[Total Marks: 9]*

**6 (a)** Interior angle between parallel North lines $= 180^\circ - 52^\circ = 128^\circ$ **[M1]**

Angles around point $B = 360^\circ - 128^\circ - 142^\circ = 90^\circ$ with no errors seen **[A1]**

**(b)** $AC = \sqrt{15^2 + 20^2}$ **[M1]**

$AC = \sqrt{625}$ **[M1]**

25 **[A1]**

**(c)** $\tan(BAC) = \dfrac{20}{15}$ **[M1]**

Angle $BAC = 53.13^\circ$ **[A1]**

Bearing $= 052^\circ + 53.13^\circ$ **[M1]**

105.1 (or 105.13 . . . ) **[A1]**

[Back to Question] *[Total Marks: 9]*

**7 (a)** $PR = \sqrt{8.5^2 + 11.2^2}$ or $\sqrt{197.69}$ **[M1]**

14.06... or 14.1 **[A1]**

**(b)** $\sin(41^\circ) = \dfrac{14.06...}{PS}$ or $PS = \dfrac{14.06...}{\sin(41^\circ)}$ **[M1]**

21.43... or 21.4 **[A1]**

**(c)** Angle $RPS = 180^\circ - 90^\circ - 41^\circ = 49^\circ$ AND distance $= 14.06... \times \sin(49^\circ)$
OR area method: $0.5 \times 21.43... \times h = 0.5 \times 14.06... \times (14.06.../\tan(41^\circ))$ **[M1]**

10.60... or 10.6 **[A1]**

[Back to Question] *[Total Marks: 6]*

**8 (a)** Sets up equation for $\triangle BCD$: $\tan(42^\circ) = \dfrac{CD}{BC}$

BC $= \dfrac{CD}{\tan(42^\circ)}$ [M1]

Sets up equation for $\triangle ACD$: $\tan(25^\circ) = \dfrac{CD}{BC + 50}$

$\text{BC} + 50 = \dfrac{CD}{\tan(25°)}$ [M1]

Equates or substitutes to eliminate $BC$: $50 = \dfrac{CD}{\tan(25°)} - \dfrac{CD}{\tan(42°)}$ **[M1]**

Factorises to solve for $CD$: $CD\left(\dfrac{1}{\tan(25°)} - \dfrac{1}{\tan(42°)}\right) = 50$ **[M1]**

48.4 (accept 48.36...) **[A1]**

**(b)** Recognises the shortest distance $d$ is the perpendicular from $B$ to $AD$, forming a right-angled triangle with hypotenuse $AB = 50$. **[M1]**

Sets up trigonometric ratio: $\sin(25°) = \dfrac{d}{50}$ **[M1]**

21.1 (accept 21.13...) **[A1]**

[Back to Question] *[Total Marks: 8]*

**9 (a)** $AC = \sqrt{8^2 + 15^2}$ **[M1]**

17 **[A1]**

**(b)** $AD^2 + 12^2 = 17^2$ (or their $AC^2$) **[M1]**

$AD = \sqrt{289 - 144} = \sqrt{145}$ **[M1]**

12.0 (or 12.04...) **[A1]**

**(c)** Area of $\triangle ADC = \dfrac{1}{2} \times 12 \times \sqrt{145} = 72.25\ldots$ **[M1]**

$\dfrac{1}{2} \times 17 \times h = 72.25\ldots$ **[M1]**

8.50 (or 8.499...) **[A1]**

**(d)** $\tan(BAC) = \dfrac{15}{8} \implies$ Angle $BAC = 61.93°$ **[M1]**

$\tan(CAD) = \dfrac{12}{\sqrt{145}} \implies$ Angle $CAD = 44.90°$ **[M1]**

Angle $DAB = 61.93° + 44.90°$ **[M1]**

106.8 (or 106.82 to 106.83) **[A1]**

[Back to Question] *[Total Marks: 12]*